

Computer Science Internal Assessment

IA Project: Student Management System

CRITERIA B - Design Overview

Table of Contents

Login Screen design.....	2
Home Screen with Menu Button Design.....	3
Student/Teacher Information Page Design.....	4
Subject Information Page Design.....	5
Marks Details Page Design.....	6
System Flow Diagram.....	7
Login Screen.....	8
Home Screen.....	8
User Profile.....	10
User Model.....	11
Student Profile.....	11
Teacher Method.....	12
Teacher Profile.....	13
Database table structure design and purpose.....	14
Database.....	14
Table: admin.....	14
Table: teacher.....	15
Table: student.....	16
Table: fee_structure.....	17
Table: student_fee.....	17
Table: subject.....	18
Table: class.....	18
Table: marks.....	19
Test Plan.....	20
Test Data.....	22
Creating Student Data.....	22
Creating Teacher Data.....	23
Marks Data.....	23
Fee Data.....	23
Subject Data.....	24
Class Data.....	24
Fee Structure.....	24

Login Screen design

LOGO (If added)	
Institute Name (If added)	
Account Type :	Drop Down Selector ▼
Username :	Text Box
Password :	Password Box

Home Screen with Menu Button Design

Teacher Details	Student Details	Class Details	Subject Details	Marks Details	Fee Details	Result	Logout

Teacher Details	Student Details	Class Details	Subject Details	Marks Details	Fee Details	Result	Logout
Add Teacher Details							
Update Teacher Details							
View Teacher Details							

Shows the Dropdown when hovered upon*

Student/Teacher Information Page Design

[illegible]

“Designs contain data as a placeholder, which will be replaced with the actual data types in the product and further documentation”

Subject Information Page Design

Teacher Details	Student Details	Class Details	Subject Details	Marks Details	Fee Details	Result	Logout
-----------------	-----------------	---------------	-----------------	---------------	-------------	--------	--------

Subject Details

Subject Code

Subject Name

Submit

Back

Marks Details Page Design

Teacher Details	Student Details	Class Details	Subject Details	Marks Details	Fee Details	Result	Logout
-----------------	-----------------	---------------	-----------------	---------------	-------------	--------	--------

Subject Details

Class Name

Student User

Student Name

Subject

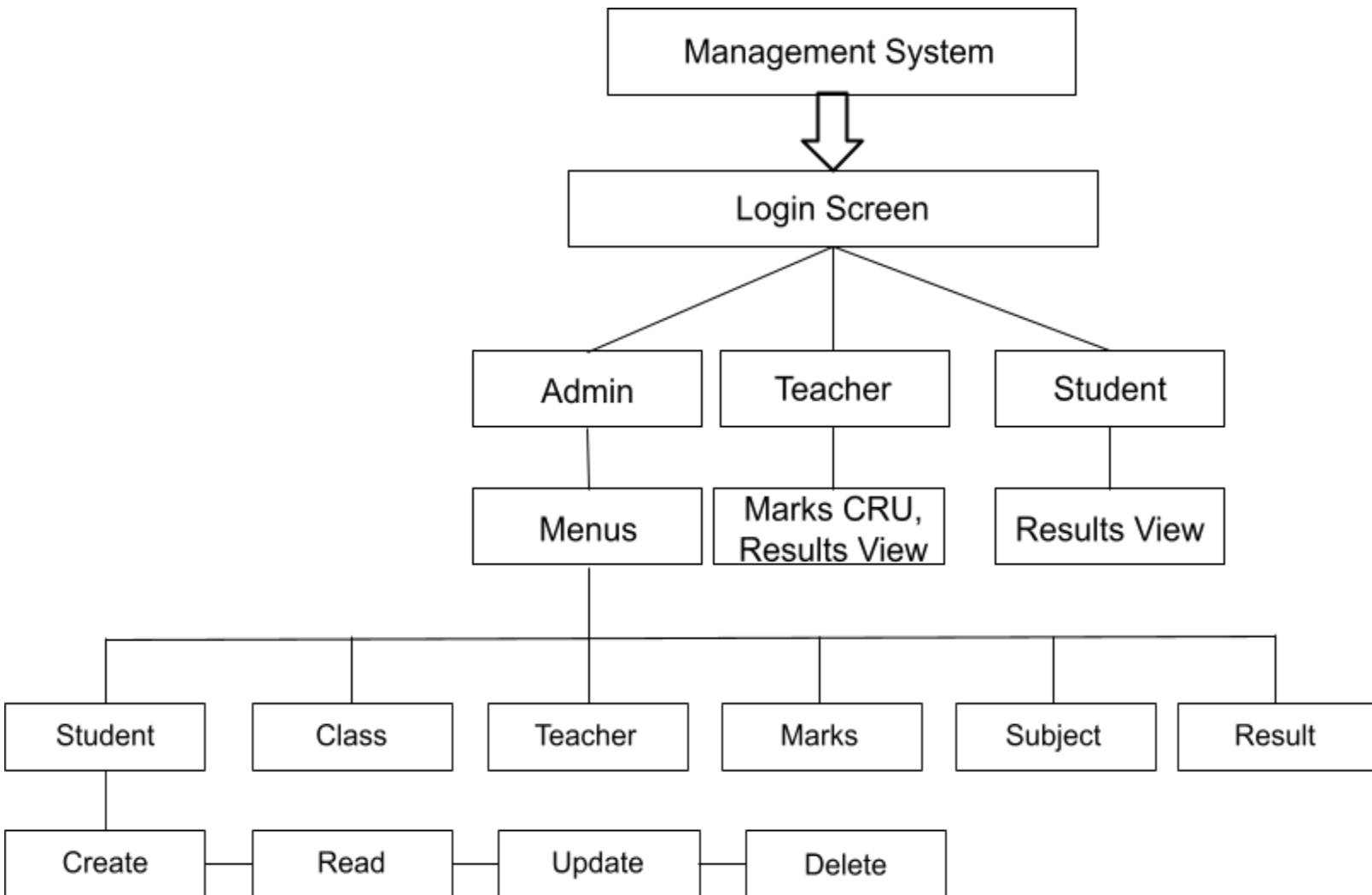
Marks

Term

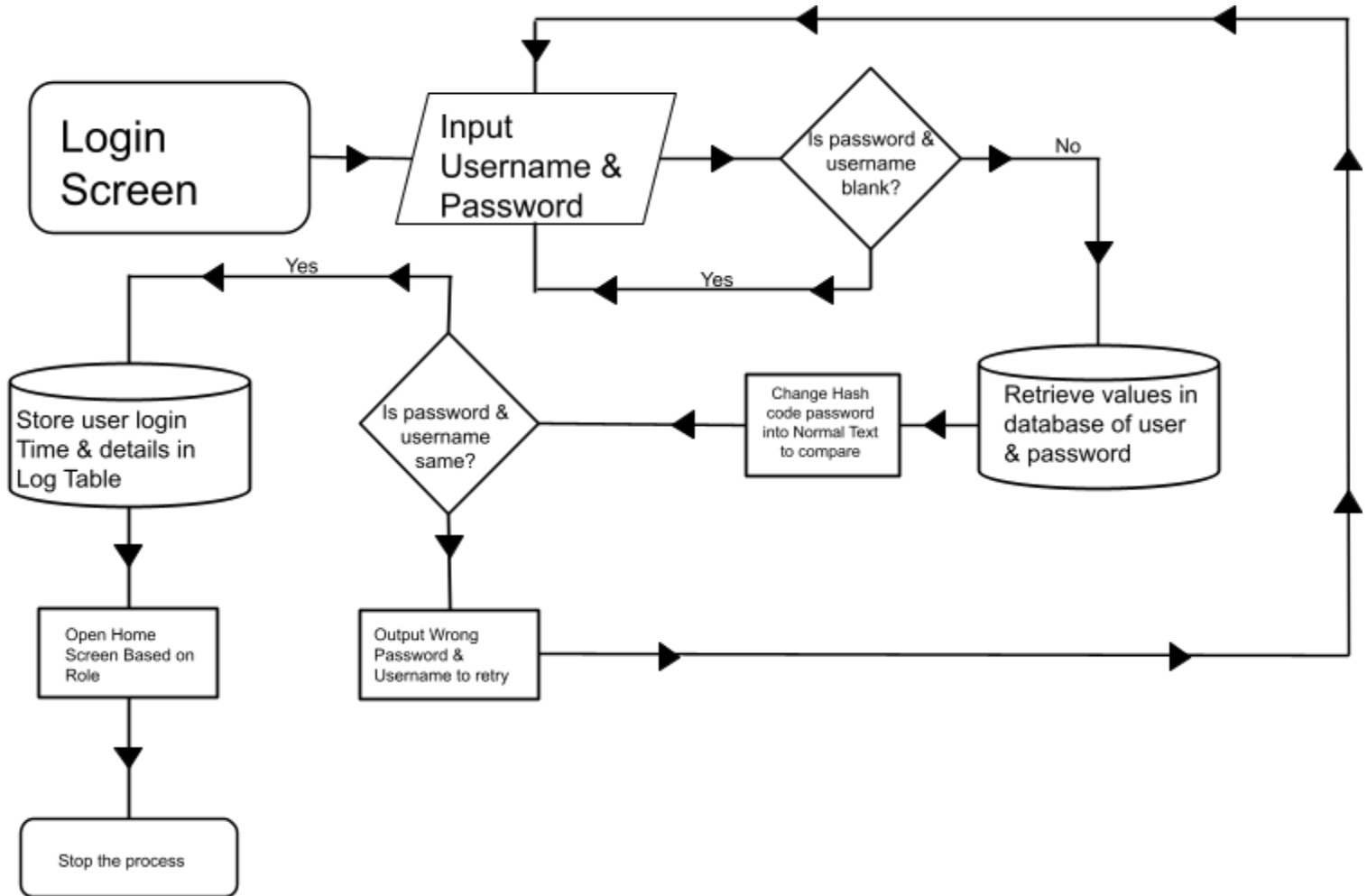
Submit

Back

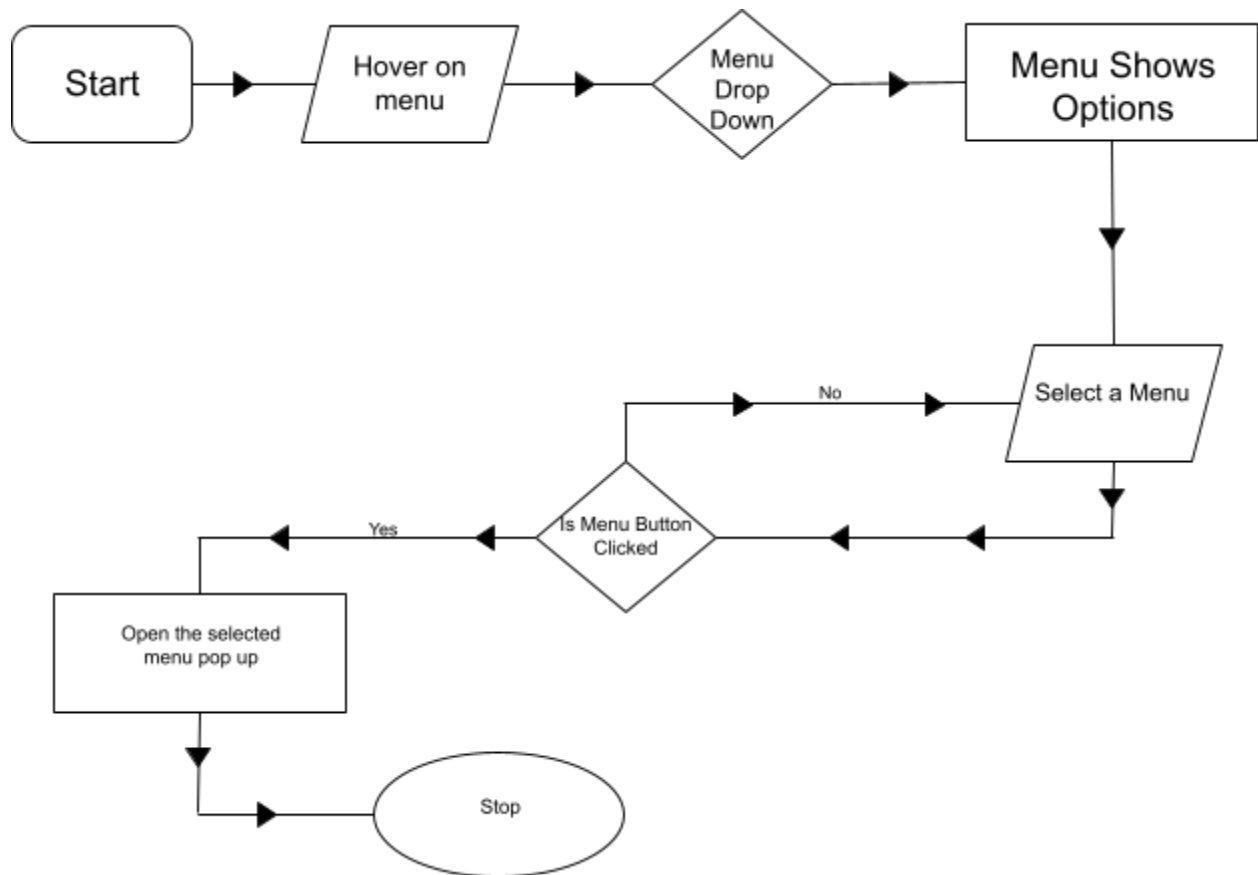
System Flow Diagram



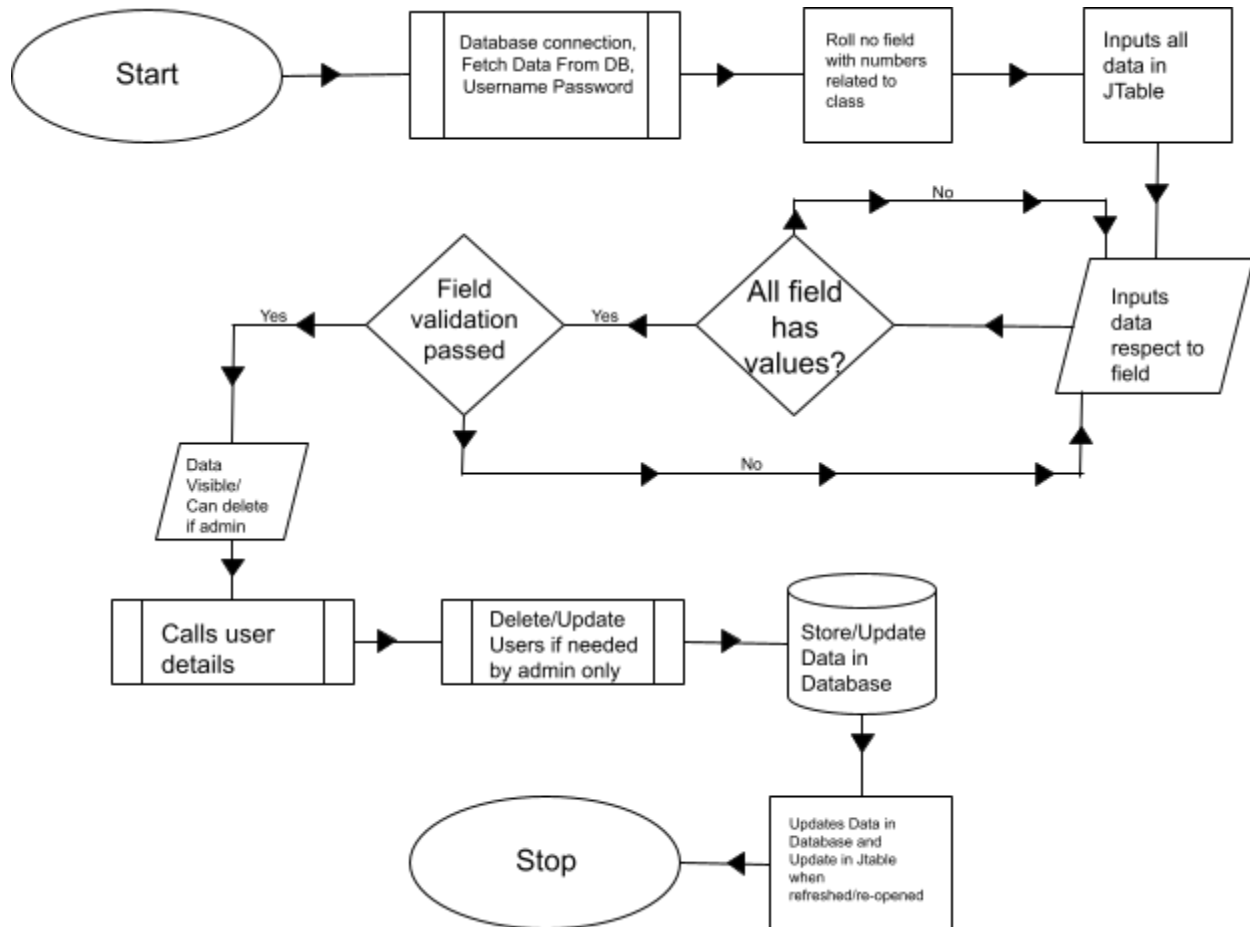
Login Screen



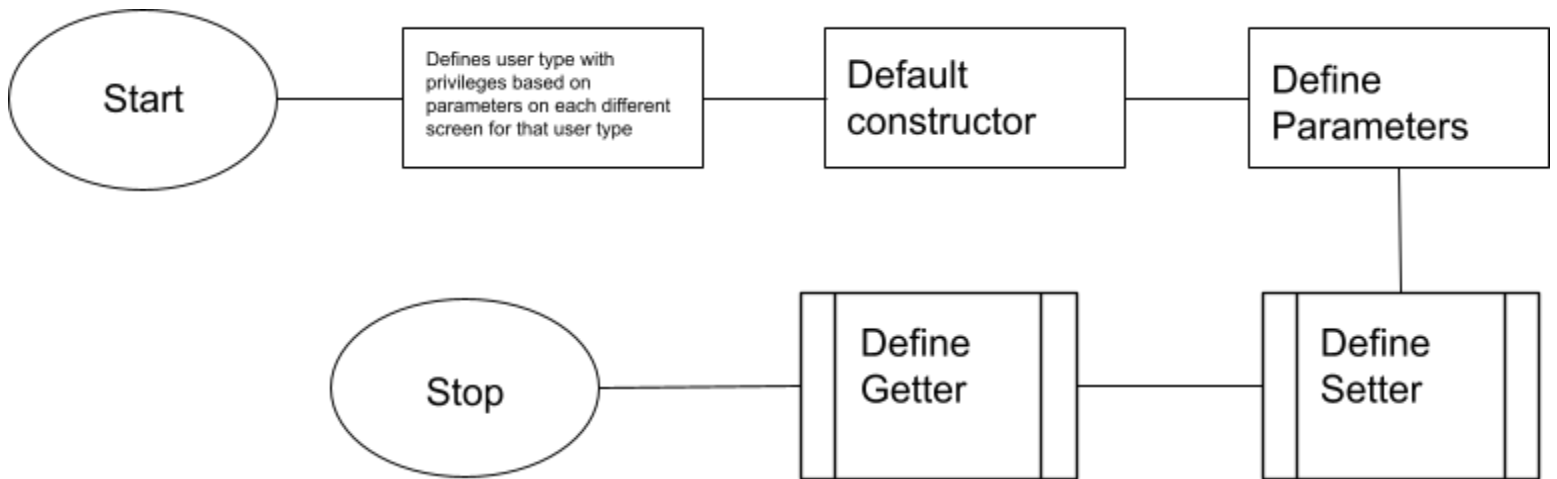
Home Screen



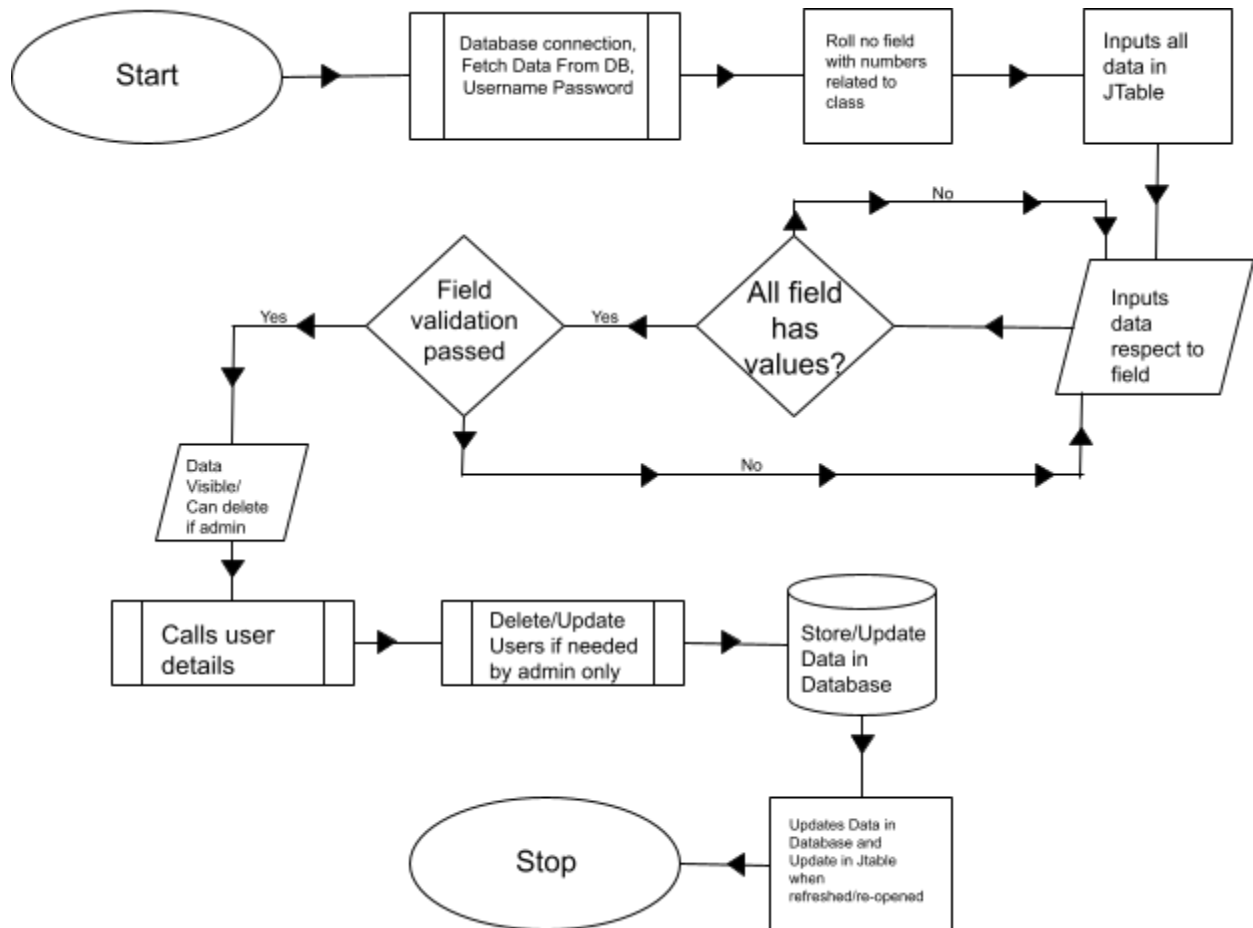
User Profile



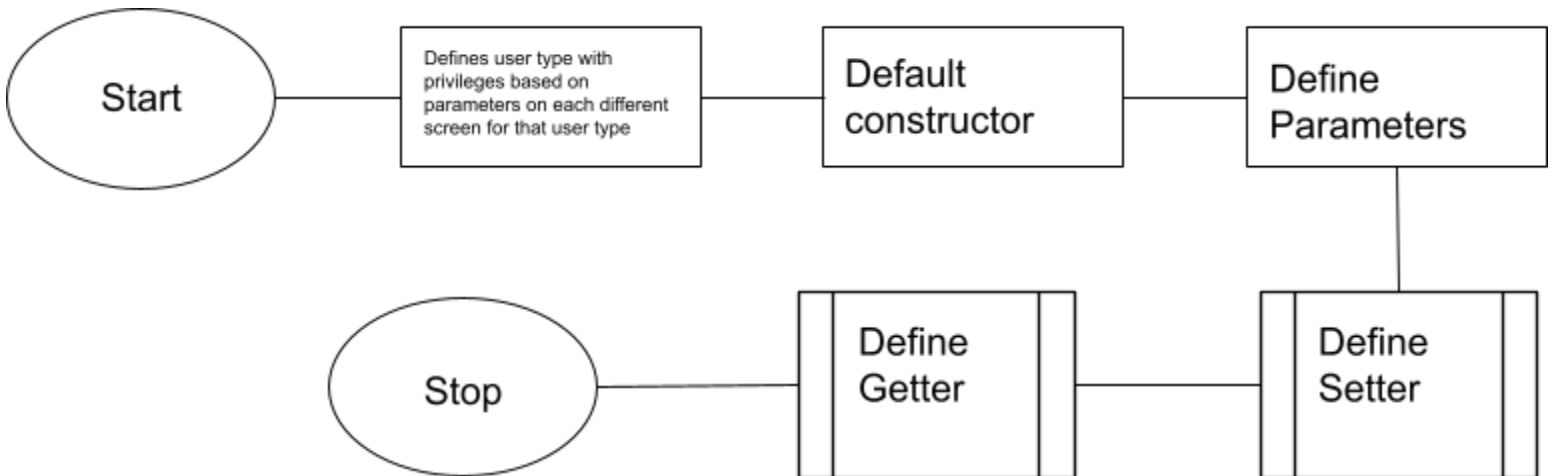
User Model



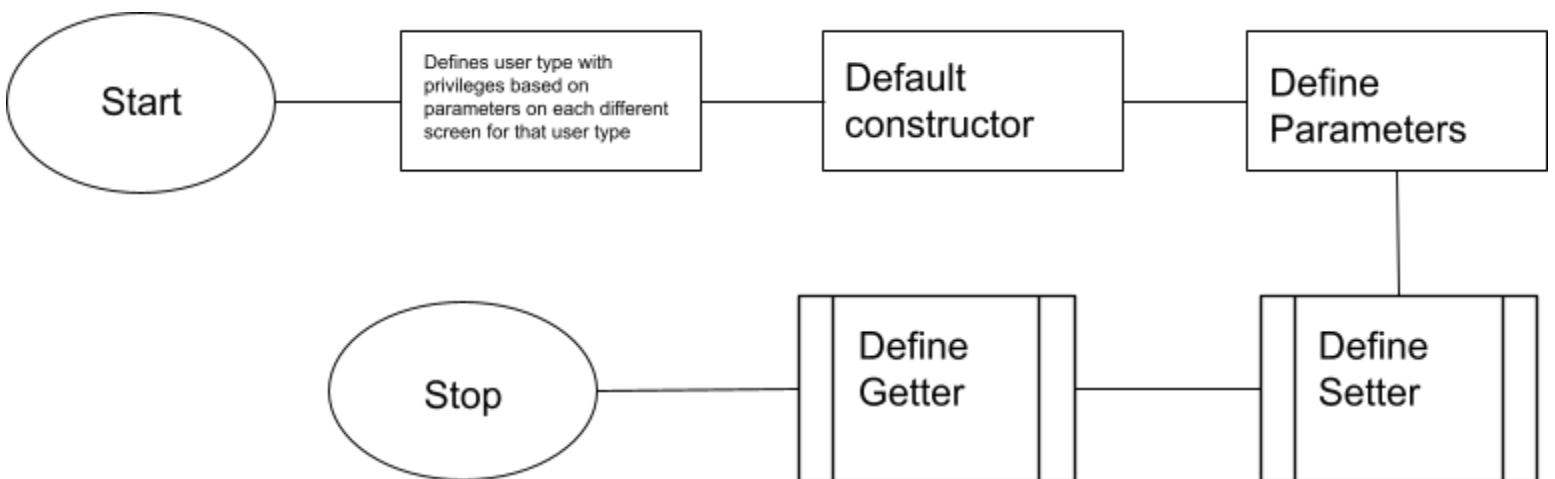
Student Profile



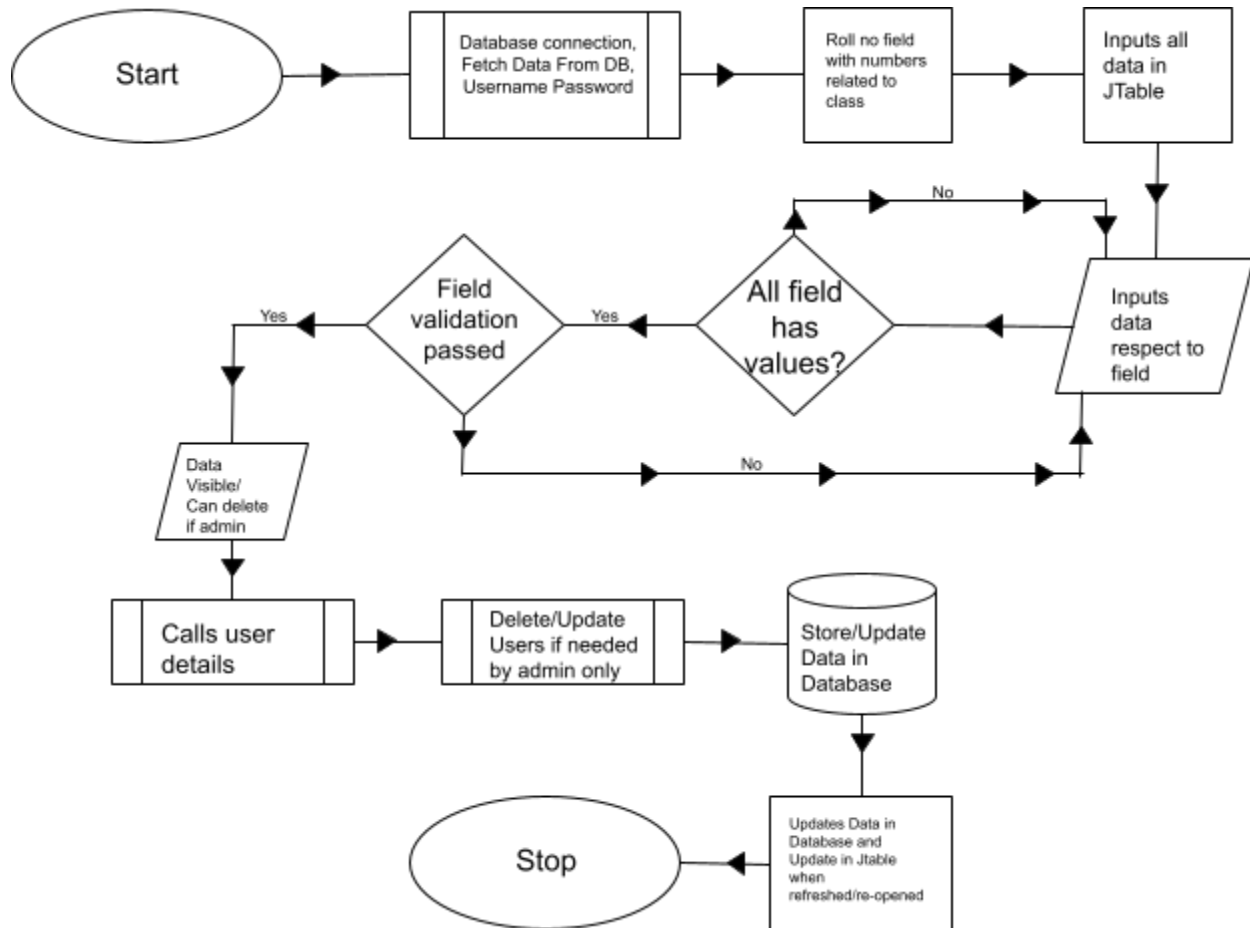
Student Method



Teacher Method



Teacher Profile



Database table structure design and purpose

Database

Table: admin

Field	Type	Field Size	Description
username	varchar	50	Stores the username of Admin
password	varchar	10	Stores the password of Admin

Table: teacher

Field	Type	Field Size	Description
tec_id	varchar	6	Holds Unique Identifier for Each Teacher Record
name	varchar	50	Holds Teacher's Name
username	varchar	50	Holds the Username of the teacher account
password	varchar	16	Holds the password of the teacher's account
email	varchar	50	Holds email ID information of the teacher
father_name	varchar	50	Holds father's name of the teacher
phone	varchar	10	Holds the phone number of the teacher
blood_group	varchar	50	Holds the blood group of the teacher
gender	varchar	50	Holds the gender information of the teacher
city	varchar	50	Holds the teacher's city details
age	varchar	50	Holds the teacher's age
teaching_exp	varchar	50	Holds the year/s of experience of the teacher
dob	varchar	50	Holds the teacher's DOB

Table: student

Field	Type	Field Size	Description
stu_id	varchar	6	Holds Unique Identifier for Each Student Record
roll_no	varchar	50	Holds the Student's Roll No data
name	varchar	50	Holds the name of the student
username	varchar	50	Holds the username of the student account
password	varchar	50	Holds the password of the student account
email	varchar	50	Holds father's name of the student
father_name	varchar	50	Holds the phone number of the student
phone	varchar	10	Holds the blood group of the student
blood	varchar	50	Holds the gender information of the teacher
gender	varchar	50	Holds the teacher's city details
city	varchar	50	Holds the teacher's age
age	varchar	50	Holds the year/s of experience of the teacher
class	varchar	10	Holds the student's DOB
dob	varchar	50	Holds the student's DOB

Table: fee_structure

Field	Type	Field Size	Description
fee_id	varchar	50	Holds unique fee ID
class_name	varchar	10	Holds the class name for which the fee is paid based on the student's grade
fee_rs	varchar	5	Holds data on how much was paid as a logger

Table: student_fee

Field	Type	Field Size	Description
fee_id	varchar	50	Holds Unique Identifier for Each Student Record
class_name	varchar	10	Holds the Student's Roll No data
username	varchar	50	Holds the username of the student account
name	varchar	50	Holds the password of the student account
email	varchar	50	Holds father's name of the student
total_fee	varchar	50	Holds the phone number of the student
submitted_fee	varchar	50	Holds the blood group of the student
status	varchar	50	Holds the gender information of the teacher
date	varchar	100	Holds the teacher's city details

Table: subject

Field	Type	Field Size	Description
subject_id	varchar	50	Holds the unique ID for any subject
subject_code	varchar	50	Holds the Subject Code for the subject (Ex, ECO, BM, GP)
subject_name	varchar	50	Holds the subject's whole name (Code: BM, Name: Business Management)

Table: class

Field	Type	Field Size	Description
cla_id	varchar	50	Holds the unique ID for the class
class_name	varchar	10	Holds the class name (ex, VIII, IX)
strength	varchar	50	Holds the data of class strength
enrolled	varchar	50	Holds the data of the students enrolled in the class

Table: marks

Field	Type	Field Size	Description
exam_id	varchar	50	Holds the unique ID for the exam
class_name	varchar	10	Holds the class name data
username	varchar	50	Holds the username of the student
name	varchar	50	Holds the students name
subject_name	varchar	50	Holds the data of which subject
marks	varchar	50	Holds the data for the marks
term	varchar	50	Holds the data for the term

Test Plan

Aspect to be tested	Test Details	Test outcome
Verify that the user interface is user-friendly and simple.	The user logs in and accesses the different areas of the dashboard, student records, and reports.	The user is knowledgeable and knows where to click; they access each feature without any confusion, and the user interface is attractive.
Validate that access to features is role-based.	The test scenario must consist of a teacher attempting to edit grades for a student, and the admin must be able to edit the user accounts	The teacher will not be able to edit grades, while the admin will have full access to control user accounts details.
Check to see if the system automatically generates unique IDs(Teacher, Student, Marks, Fees)	Add multiple records	Each record has a unique ID, without duplicates in it's data class.
Test the addition data without credentials (Teacher, Student, Class, Subject, Marks, Fees)	Try to create information in the system without credentials	The system will prompt the user to enter the required credentials
Now test the system for creating details	Enter the information from test data and conform data is created and that it is stored in the system database.	New data is created and stored in the database
Test if the password is encrypted during account creation (Teacher, Student)	Check if the passwords are encrypted and inserted into the database during user creation	The password is encrypted and securely inserted into the database holding student account details
Test and confirm that the system deletes the data from the database	Enter the information and click delete.	Confirm that it has been deleted from the database.
Test and confirm the Fee Function of the System	Test if it can add records of paid, pending and due into the history with data, also	The records are successfully inserted into the database with proper Data and is also

	with the complex logic working with the mathematical logics of fees pending and paid amounts	visible in the View Fee Details Panel.
Test and confirm the Marks Function of the system	Test if the system can add record of marks of individual students into the system	The records are successfully inserted into the database with proper data and is also visible in the View Marks details.
Test and conform the Result Function of the system	Test if the system can calculate and compile the results of the students of each student properly with total percentage scores from all terms which hae been added	The records are successfully inserted into the database with proper data and is also visible in the results panel compiled individually with all terms and marks details along with proper percentage scores.

Test Data

Creating Student Data

Name	username	Password	Grade	email	DOB	father name	Phone number	blood	gender	age	City
Sharvil Gala	student81	student123	VIII	student81@gmail.com	1/1/2012	Manan Gala	123456789	O+	male	14	Kolhapur
Praful Nahata	student91	student123	IX	student91@gmail.com	1/1/2011	Khushtahata	789456123	A+	male	15	Pune
Arsha Kadam	student101	student123	X	student101@gmail.com	1/1/2010	Atharv Kadam	654789321	A-	female	16	Karad
Nidhi Munde	studentib11	student123	IBD P-1	nidhi.munde@gmail.com	1/1/2009	Sai Munde	456789321	O+	female	17	Lathur
Sidhi Gosavi	studentib21	student123	IBD P-2	sidhi.gosavi@gmail.com	1/1/2008	Shubham K Gosavi	423651798	B+	female	18	Mumbai
Samrudhi Jawar	studentas1	student123	AS-Level	samrudhi.jawar@gmail.com	1/1/2009	Shlok Jawar	124785963	A+	female	17	Hydrabad
Nishi Jain	studenta1	student123	A-Level	nishi.jain@gmail.com	1/1/2008	Tanesh Jain	124698538	O+	female	18	Bangalore

Creating Teacher Data

name	username	Password	Teacher Exp	email	DOB	Phone number	eblood	gender	age	City
Mahi Kumari	teacher1	teacher123	2	mahi.kuamri@gmail.com	1/1/2000	987654321	B+	female	26	Kolhapur
Darshan Jain	teacher2	teacher123	1	darshan.j@gmail.com	1/1/1999	456123789	O+	female	27	Sangli

Marks Data

Class	Student_username	Subject	Marks	Term
VIII	student81	Economics	80	First
VIII	student81	Economics	85	Second
IX	student91	Bussiness Management	83	First
IX	student91	Bussiness Management	89	Second

Fee Data

Grade	Username	Amount	Status
X	student101	2000	Completed
X	student101	1000	Due
X	student101	1000	Completed

Check the next possible to add more

Subject Data

Subject name	Subject Code
Economics	ECO
Business Management	BM
Global Politics	GP

Class Data

Class Name	Class Strength
VIII	5
IX	5
X	5
IBDP-1	5
IBDP-2	5
AS-Level	5
A-Level	5

Fee Structure

Class Name	Fee/Year (Rs)
VIII	11000
IX	12000
X	20000
IBDP-1	600000
IBDP-2	700000
AS-Level	140000
A-Level	150000